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Modeling Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

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Abstract

It has been observed that patients in intensive care are likely to suffer from hyperglycaemia as a result of fluctuating insulin resistance. Controlling blood glucose from insulin and nutrition protocols stemming from hourly glucose measurements has proved to be difficult and not sophisticated enough to consider the dynamic health of the patient. Implementation of tight glucose control has been motivated by the effectiveness of computational estimation of insulin-glucose states in synthetic and real patients by using mechanistic models such as the Intensive Control and Insulin-Nutrition-Glucose (ICING) model. This thesis sought to improve the design of synthetic patients and the application of Bayesian filters for estimation. Better simulation and design of tight glucose controls is crucial to implement improved treatment of patients with critical illnesses. An analytical framework to sample synthetic patient trajectories was proposed that utilized distributions of insulin and enteral/parenteral inputs given to 10702 patients in the Karolinska University Hospital. An estimation of unknown insulin-glucose states from 250 unique patients was compared by analyzing the mean squared error (MSE) and the run time of the estimates in the extended Kalman filter, and a bootstrap particle filter of various Monte Carlo samples for a patient over 12 hours. The extended Kalman filter had an average MSE of 3.46 and a run time of 0.371 seconds, while a particle filter with 1000 Monte Carlo samples achieved an average MSE of 0.926 and a run time of 8.32 seconds. Estimation of an additional parameter of constant insulin sensitivity through state augmentation in another set of 250 synthetic patients resulted in the extended Kalman filter having a MSE of 5.62×10^{-7} , and the particle filter of 1000 particles with a MSE of 1.49×10^{-7} . For a changing insulin sensitivity, both filters had an increase of MSE of one order of magnitude. Given utilizing the ICING model with blood glucose measurements, a particle filter with 1000 particles can estimate blood glucose and constant insulin sensitivity in critically ill patients with a reasonable run time. Further investigation into capturing time-varying insulin sensitivity is needed to better model patient dynamics.

Keywords

Bayesian filter, Blood glucose, Critically ill patients, Insulin resistance, Insulin-glucose dynamics, Insulin sensitivity, Model-based control, Tight glycemic control

Sammanfattning

Det har observerats att patienter inom intensivvården sannolikt drabbas av hyperglykemi till följd av fluktuerande insulinresistens. Att kontrollera blodsockret med insulin- och nutritionsprotokoll som härrör från timvisa glukosmätningar har visat sig vara svårt och inte tillräckligt sofistikerat för att beakta patientens dynamiska hälsotillstånd. Implementeringen av strikt glukoskontroll har motiverats av effektiviteten hos beräkningsbaserad skattning av insulin-glukostillstånd hos syntetiska och verkliga patienter genom användning av mekanistiska modeller såsom Intensive Control and Insulin-Nutrition-Glucose (**ICING**)-modellen. Denna avhandling syftade till att förbättra utformningen av syntetiska patienter och tillämpningen av Bayesianska filter för skattning. Bättre simulering och utformning av strikt glukoskontroll är avgörande för att implementera förbättrad behandling av patienter med kritiska sjukdomstillstånd. Ett analytiskt ramverk för att sampla syntetiska patientförlopp föreslogs, vilket använde fördelningar av insulin- och enterala/parenterala tillförselmängder som gavs till 10702 patienter vid Karolinska University Hospital. En skattning av okända insulin-glukostillstånd från 250 unika patienter jämfördes genom analys av medelkvadratfelet (**MSE**) och körtiden för skattningarna i det utvidgade Kalmanfiltret samt ett bootstrap-partikelfilter med olika antal Monte Carlo-samplingar för en patient under 12 timmar. Det utvidgade Kalmanfiltret hade ett genomsnittligt **MSE** på 3.46 och en körtid på 0.371 sekunder, medan ett partikelfilter med 1000 Monte Carlo-samplingar uppnådde ett genomsnittligt **MSE** på 0.926 och en körtid på 8.32 sekunder. Skattning av ytterligare en parameter för konstant insulinkänslighet genom tillståndsaugmentering hos ytterligare 250 syntetiska patienter resulterade i att det utvidgade Kalmanfiltret hade ett **MSE** på 5.62×10^{-7} , och partikelfiltret med 1000 partiklar hade ett **MSE** på 1.49×10^{-7} . Vid varierande insulinkänslighet hade båda filtren en ökning av **MSE** med en storleksordning. Givet användningen av **ICING**-modellen tillsammans med blodsockermätningar kan ett partikelfilter med 1000 partiklar uppskatta blodsocker och konstant insulinkänslighet hos kritiskt sjuka patienter med en rimlig körtid. Ytterligare undersökning av hur tidsvarierande insulinkänslighet kan fångas är nödvändig för att bättre modellera patientdynamiken.

Nyckelord

Bayesiskt filter, Blodglukos, Insulin-glukos dynamiker, Insulinkänslighet, Insulinresisten, Kritiskt sjuka patienter, Modellbaserade kontrollera, Tight

glycemic control

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Acronyms

ICING	Intensive Control Insulin-Nutrition-Glucose
IPI	Integral-based Parameter Identification
MSE	Mean squared error

Chapter 1

Introduction

Patients in the intensive care unit suffering from critical illnesses (e.g. acute myocardial infarction, sepsis, stroke) have been widely documented to undergo hyperglycaemia, even those without a prior diagnosis of diabetes [1]. Acute stress observed in critical illnesses often induces physiological changes, as the secretion of stress hormones results in an inflammatory response associated with insulin-resistance [2] [3] [4] [5] [6] [7]. While there is inconclusive evidence pertaining to the causality of stress hyperglycaemia and adverse outcomes, the onset of mortality is far greater in patients with no reported history of hyperglycaemia [1] [3]. Sustaining glucose concentration under 6.106 mmol/L through insulin infusion has demonstrated a reduction of mortality and morbidity, compared to less-intensive methods of glycaemic control [8], at the subsequent cost of an increase of reported incidences of hypoglycaemia [9]. Further review of intensive insulin therapy and tight glucose control has identified discrepancies in the administration of glycaemic control, mostly corresponding to the risk of moderate to severe hypoglycaemia and glycaemic variability in a critically ill patient is equally detrimental as stress hyperglycaemia, with an increase of mortality and health complications observed [9] [10] [11].

1.1 Background

Clinicians now aim to maintain blood glucose in a range of 6 mmol/L to 10 mmol/L to account for adverse outcomes in hyperglycaemia and hypoglycaemia. Ad hoc sliding protocols have been utilized in the intensive care unit to provide clinicians with the necessary nutrition and insulin doses to control blood glucose levels based on hourly glucose measurements [12]

[13]. Such methods are often difficult to implement in practice and result in minimal improvements in preventing mortality [14]. The implementation of tight glucose controllers avoids tedious procedures followed in sliding protocols by estimating insulin-glucose states on given insulin and nutrition inputs, with observed improvements in constraining blood glucose to a safe range [15]. The implementation of model-based tight glucose control methods relies on the formulation of insulin-glucose dynamics. Mechanistic insulin-glucose models have been developed with an extensive physiological foundation, and have been validated for glycaemic control in clinical settings [14] [15] [16] [17] [18]. The insulin-glucose models derived are nonlinear, which requires more advanced numerical methods to properly estimate the current and future insulin-glucose states in patients. In practice, the estimated states from the insulin-glucose models are affected by process noise from unmodeled states (and/or parameters) in the model formulation and from noisy measurements. As a direct result, the insulin-glucose states in practice struggle to be estimated deterministically. To accommodate modeling of stochastic processes, statistical methods are needed to optimally estimate the states in mechanistic insulin-glucose models.

1.2 Problem

Patients suffering with differing critical illnesses are observed with varying magnitudes of insulin resistance, preventing a one-size-fits-all tight glucose control protocol to restrict glycaemic levels. Model-based tight glucose control has the ability to adapt to a given patient's condition, but there currently exist gaps in research for in-depth analysis of statistical modeling of insulin-glucose states. Previous studies have employed Bayesian filtering methods for this task [19] [20] and integral-based approximation to estimate unknown insulin-glucose model parameters [18]. These studies lack systematic design of the synthetic patient cohort and practical implementation, with studies containing constant nutrition and insulin inputs, or with known initial insulin-glucose dynamics. The gap in research in analyzing various Bayesian filters with representative synthetic patients hinders gauging how well tight glucose controllers work in practice with vulnerable patients.

1.3 Purpose

This thesis aims to improve the current research and implementation of tight blood glucose control. Researchers studying insulin-glucose systems in critically ill patients will benefit from a cohesive and reproducible methodology to accommodate their own research goals. Better glycaemic control will result in clinicians being able to adjust nutrition and insulin doses from the prediction of insulin-glucose states in between hourly measurements. Consequently, giving clinicians crucial time to better prevent mortality in patients susceptible to hypoglycaemia or hyperglycaemia.

1.4 Goals

The overall goals of this project has been divided into the following sub-goals:

1. Design a framework to generate synthetic patient trajectories
2. Compare performance of various filters in estimation of insulin-glucose states
3. Estimate unknown insulin-glucose model parameters in patients

1.5 Research Methods

The modeling of the insulin-glucose dynamics found in critically ill patients will be implemented on a synthetic patient cohort. Using synthetic patients allows the researcher to manually control the insulin-glucose model parameters, which directly improves the evaluation of the outcome of the insulin-glucose control. Previous work will serve as the benchmark to compare how the parameterization of our method performs in estimating insulin-glucose dynamics of synthetic patients. The chosen filter design will then be used to estimate additional dynamics not mathematically represented in mechanistic insulin-glucose models. Fitting metrics and computational complexity of the estimators will be utilized to draw inferences about the performance.

1.6 Delimitations

This study will consist of synthetic patient trajectories. Using synthetic patients is important for maintaining internal validity in the experimental process, as it verifies the observed outcome of the experiment is due to the manipulation of insulin-glucose model parameters rather than uncontrolled variables. The expense of using synthetic patients can be identified in the reduced generalizability and external validity of this experiment, since synthetic patients are reflections of model assumptions. While this is a limitation, this study will focus on the foundation of the modeling of insulin-glucose dynamics, and future work can replicate the study with real clinical data.

1.7 Structure of the Thesis

Chapter 2 will introduce the existing modeling of insulin-glucose dynamics used, followed by an overview of discretization, state estimation, and related work in parameter estimation. In Chapter 3, the proposed research process used to answer the research questions of this study will be documented and motivated. The results and the subsequent discussion will be presented in Chapter 4. The key findings and future work of the experiment will be summarized in Chapter 5.

Chapter 2

Background

The background will introduce the theoretical concepts and prior work relevant to the thesis. Beginning with a description of the system physiology and the Intensive Control Insulin-Nutrition-Glucose (**ICING**) model. Followed by presenting how state estimation methods can be applied to estimate insulin-glucose states optimally. The chapter will then conclude with practical parameter estimation approaches to predict unknown parameters.

2.1 Physiology of Glucoregulation

To model glucoregulation in critically ill patients, it is necessary to understand how the physiological mechanisms are interrelated. Glucoregulation in the human body relies on the function of the pancreas. When there is a decrease in blood glucose, the alpha cells in the pancreas produce the hormone glucagon, that binds to liver cells. Resulting in the conversion of the stored glycogen in the liver to glucose, which is secreted in the bloodstream. In the case of an increase of blood glucose, such as the intake of a meal, beta cells in the pancreas will release insulin. The kidneys will filter the produced insulin, and the liver will be signaled to convert glucose into glycogen. Thus, decreasing the total amount of blood glucose.

In cases of hyperglycaemia in the intensive care unit, a patient may be given exogenous insulin to maintain a patient's blood glucose in a safe range. However, increased insulin resistance has been observed in critically-ill patients. As a result, the body has a reduced reaction to insulin, consequently, impairing glucoregulation in the body. Given the complexities of the mechanisms, a model can provide crucial insight into critically ill patients.

2.2 Intensive Control Insulin-Nutrition-Glucose Model

The **ICING** model presented by Lin et al. [17] captures the insulin-glucose dynamics in critically ill patients. The model is constructed as a fusion of two existing models validated for patients in the intensive care unit [14] [16], containing the following ensemble of differential equations

$$\dot{G} = -p_G G(t) - S_I(t) G(t) \frac{Q(t)}{1 + \alpha_G Q(t)} + \frac{P(t) + E_G - C_G}{V_G} \quad (2.1a)$$

$$\dot{Q} = n_I(I(t) - Q(t)) - n_C \frac{Q(t)}{1 + \alpha_G Q(t)} \quad (2.1b)$$

$$\begin{aligned} \dot{I} = & -n_K I(t) - n_L \frac{I(t)}{1 + \alpha_I I(t)} - n_I(I(t) - Q(t)) + \frac{u_{ex}(t)}{V_I} \\ & + \frac{(1 - x_L)u_{en}(t)}{V_I} \end{aligned} \quad (2.1c)$$

$$\dot{P}_1 = -d_1 P_1 + D(t) \quad (2.1d)$$

$$\dot{P}_2 = -\min(d_2 P_2, P_{max}) + d_1 P_1 \quad (2.1e)$$

$$u_{en}(t) = k_1 \exp\left(-\frac{k_2}{k_3} I(t)\right), \quad \text{when C-peptide data is not available} \quad (2.1f)$$

$$P(t) = \min(d_2 P_2, P_{max}) + P_N(t). \quad (2.1g)$$

Equation (2.1a) is the independent insulin glucose removal, with G being the total blood glucose concentration (mmol/L). The whole-body insulin sensitivity $S_I(t)$ gives insight into the metabolic condition of the patient and is a time-varying dynamic. The insulin pharmacodynamics are shown in Equations (2.1b) and (2.1c), with Q and I being the interstitial insulin concentration (mU/L) and the plasma insulin concentration (mU/L) respectively. The exogenous insulin input is $u_{ex}(t)$ (mU/min), and the endogenous insulin production $u_{en}(t)$ (mU/min) can be measured from C-Peptide, but if measurements are not feasible, the kinetics can model as such in Equation (2.1f). The gastric absorption of glucose is defined in Equations (2.1d) to (2.1g). The dynamics P_1 and P_2 describe the glucose in the stomach and the gut (mmol), with $P(t)$ representing the glucose appearance in the blood (mmol/min). Any enteral food intake given to the patient is accounted for by $D(t)$ (mmol/min), and additional parenteral glucose is described by $P_N(t)$ (mmol/min).

The **ICING** model also involves several parameters alongside the previously defined dynamics, which are detailed in Table 2.1. The majority of the patient model parameters are assigned as general population constants observed from clinical trials, and are validated in relevant insulin-glucose control parameter sensitivity studies [18] [15] [21]. The identification of the remaining parameters was estimated with a data-driven approach by fitting the patient parameters to the observed dynamics in 173 patients, and by assessing the impact of prediction errors [21]. Estimating the endogenous glucose removal p_G and the suppression of endogenous glucose produce E_G constants involved assigning all other patient parameters and the insulin sensitivity to constants, then starting a grid search that covers $p_G = 0.001, 0.002, \dots, 0.1(\text{min}^{-1})$ and $E_G = 0.0, 0.1, \dots, 3.5 (\text{mmol/min})$. The joint p_G and E_G value that achieved the lowest fitting and prediction error for blood glucose in Equation (2.1a) was chosen. Ultimately, that was $p_G = 0.006 (\text{min}^{-1})$ and $E_G = 1.16 (\text{mmol/min})$, while both falling in the ranges observed physiologically. The glucose distribution rate between the stomach and the gut n_I , and cellular insulin clearance rate in the interstitium were estimated similarly by fitting to Q in the Equation (2.1b), a grid search covers a range of $n_I = n_C = 0.0001, \dots, 0.02 (\text{min}^{-1})$, with a value of $n_I = 0.003 (\text{min}^{-1})$ resulting in the best predictive performance.

It should be noted that in the **ICING** model, the nonlinear dynamics are defined in continuous-time. Discretization of the model is necessary as in the following section, the estimation methods presented operate within discrete time, noted as k . If we define the discrete time step as $t_k = t_{k-1} + \Delta t$, an example state $F(t)$ can be discretized according to the Euler method

$$\begin{aligned}\dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta t)}{\Delta t} \\ F_k &\approx F_{k-1} + \Delta t \dot{F}.\end{aligned}\tag{2.2}$$

The error of this numerical approximation of differential equations is bounded by the step size Δt . As a result, larger step sizes will introduce greater approximation errors. Thus, a relatively small step size is necessary to prevent errors from propagating in the estimates.

A probabilistic representation of the **ICING** model will be utilized in this thesis for Bayesian filtering. The states in Equations (2.1a) to (2.1g), will be described as a Bayesian inference from the previous states and inputs. Modeling the **ICING** equations as a random process will be done to model the uncertainty of the states. The specific conditional probability distribution

Table 2.1: Assigned parameters in the **ICING** model. The parameters shaded in gray are derived from fitting methods.

Parameter	Notation	Value
Patient endogenous glucose removal	p_G (min^{-1})	0.006
Saturation of insulin-stimulated glucose removal	α_G (L/mU)	0.0154
Suppression of endogenous glucose production	E_G (mmol/min)	1.16
Central nervous system uptake	C_G (mmol/L)	0.3
Glucose distribution volume	V_G (L)	13.3
Glucose distribution rate between the stomach and the gut	n_I (min^{-1})	0.003
Cellular insulin clearance rate in the interstitium	n_C (min^{-1})	$= n_I$
Kidney clearance rate of insulin from plasma	n_K (min^{-1})	0.0542
Liver clearance rate of insulin from plasma	n_L (min^{-1})	0.01578
Saturation of plasma insulin disappearance	α_I (L/mU)	0.0017
Insulin distribution volume	V_I (L)	3.15
Fraction of endogenous insulin hepatic extraction	x_L	0.67
Maximal glucose out flux from the gut	P_{max} (mmol/min)	6.11
Base endogenous insulin production rate	k_1 (mU/min)	45.7
Transport rates between the stomach and the gut	d_1 (min^{-1})	0.0347
	d_2 (min^{-1})	0.0069
Exponential suppression constants	k_2	1.5
	k_3	1000

will be dependent on the type of Bayesian filter, as seen in the next section.

2.3 State Estimation

Within the scope of this thesis, discrete-time dynamical systems will be modeled in the state-space representation. A state-space model consists of system variables that describe the state of the system, inputs, and measurements. The states are typically formulated as linear or nonlinear equations. A probabilistic representation of the discrete-time state-space can be represented as such

$$\begin{aligned} x_k &\sim p(x_k \mid x_{k-1}, u_{k-1}), \\ y_k &\sim p(y_k \mid x_k), \end{aligned} \tag{2.3}$$

where $p(x_k | x_{k-1}, u_k)$ and $p(y_k | x_k)$ are the stochastic discrete-time dynamic and measurement models, with $x_k \in \mathbb{R}^n$ representing the states, $u_k \in \mathbb{R}^p$ are the inputs, and $y_k \in \mathbb{R}^m$ are the measurements.

Any model has intrinsic approximation errors, and the errors should be taken into account. The approximation error for the states can be modeled as a random process, and artificial process noise can be added into x_k at each time step. This step is important to accommodate errors due to unmodeled dynamics. Choices in the specifics regarding the modeling of process noise of insulin-glucose states will be discussed in Chapter 3.

The objective of state estimators can be described as estimating the distribution of the state x_k , given all the measurements up to the current sample $p(x_k | y_{1:k})$. In the case of assuming the model is linear and Gaussian, the states and measurements can be written as the difference equations

$$\begin{aligned} x_k &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + q_{k-1}, \\ y_k &= Hx_k + r_k, \end{aligned} \quad (2.4)$$

where $q_{k-1} \sim \mathcal{N}(0, Q_{k-1})$ is the process noise, $r_k \sim \mathcal{N}(0, R_k)$ is the measurement noise, we can rewrite the Equation (2.3) as the following

$$\begin{aligned} p(x_k | x_{k-1}, u_{k-1}) &= \mathcal{N}(x_k | A_{k-1}x_{k-1} + B_{k-1}u_{k-1}, Q_{k-1}), \\ p(y_k | x_k) &= \mathcal{N}(y_k | H_k x_k, R_k). \end{aligned} \quad (2.5)$$

The closed-form solution of $p(x_k | y_{1:k})$ for linear and Gaussian dynamic and measurement functions is known as the Kalman filter, with a full derivation available from Särkkä and Svensson [22]. The filter contains two key parts: the *prediction step* and the *update step*.

1. Prediction step

$$\begin{aligned} m_k^- &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1}, \\ P_k^- &= A_{k-1}P_{k-1}A_{k-1}^\top + Q_{k-1}. \end{aligned} \quad (2.6)$$

2. Update step

$$\begin{aligned} v_k &= y_k - H_k m_k^-, \\ S_k &= H_k P_k^- H_k^\top + R_k, \\ K_k &= P_k^- H_k^\top S_k^{-1}, \\ m_k &= m_k^- + K_k v_k, \\ P_k &= P_k^- - K_k S_k K_k^\top. \end{aligned} \quad (2.7)$$

The Kalman filter is recursive, as the filter starts with an estimate of the initial distribution of the state m_0^- and the covariance P_0^- , and iterates sequentially over discrete time steps k . In practice, the Kalman filter is often a great starting point for an estimator because the assumptions of linearity and Gaussianity often hold in many contexts, but as Equations (2.1a) to (2.1e) are nonlinear, other methods must be considered.

2.3.1 Extended Kalman Filter

The extended Kalman filter can accommodate the nonlinearities in the **ICING** model by approximating the dynamic and measurement models as a linear transformation of the states, inputs, and measurements. The structure of the extended Kalman filter remains the same as the Equations (2.6) and (2.7), but changes are seen in the matrices of the dynamic and measurement models $f(\cdot)$ and $h(\cdot)$. Assuming that the dynamic and measurement models are Gaussian, the prediction and update steps [22] can be defined as

1. Prediction step

$$\begin{aligned} m_k^- &= f(m_{k-1}, u_{k-1}), \\ P_k^- &= F_x(m_{k-1})P_{k-1}F_x^\top(m_{k-1}) + F_q(m_{k-1})Q_{k-1}F_q^\top(m_{k-1}). \end{aligned} \quad (2.8)$$

2. Update step

$$\begin{aligned} v_k &= y_k - h(m_k^-), \\ S_k &= H_x(m_k^-)P_k^-H_x^\top(m_k^-) + H_r(m_k^-)R_kH_r^\top(m_k^-), \\ K_k &= P_k^-H_k^\top(m_k^-)S_k^{-1}, \\ m_k &= m_k^- + K_kv_k, \\ P_k &= P_k^- - K_kS_kK_k^\top, \end{aligned} \quad (2.9)$$

the Jacobian matrices $F_x(m_{k-1})$, $F_q(m_{k-1})$, $H_x(m_k^-)$, and $H_r(m_k^-)$ are the Taylor series linear approximation of the dynamic and measurement models

$$\begin{aligned}
 [F_x(m_{k-1})]_{j,j'} &= \left. \frac{\partial f_j(x, q)}{\partial x_{j'}} \right|_{x=m_{k-1}, q=0}, \\
 [F_q(m_{k-1})]_{j,j'} &= \left. \frac{\partial f_j(x, q)}{\partial q_{j'}} \right|_{x=m_{k-1}, q=0}, \\
 [H_x(m_k^-)]_{j,j'} &= \left. \frac{\partial h_j(x, r)}{\partial x_{j'}} \right|_{x=m_k^-, r=0}, \\
 [H_r(m_k^-)]_{j,j'} &= \left. \frac{\partial h_j(x, r)}{\partial r_{j'}} \right|_{x=m_k^-, r=0}.
 \end{aligned} \tag{2.10}$$

The intrinsic bottleneck of the extended Kalman filter can be ascribed to the Gaussian formulation of the state update equations. Any deviations from Gaussianity in the posterior will directly degrade the estimation capabilities of the estimator. Another type of state estimation method is the particle filter, which can accommodate nonlinear and nongaussian dynamic and measurement models by using point estimates from a Monte Carlo approximation of the posterior $p(x_k | y_{1:k})$.

2.3.2 Bootstrap Particle Filter

For state-space modeling, the dynamic and measurement models are run sequentially such that x_k and y_k are represented as a Markov Chain seen in Equation (2.3). In the Kalman filter case, such dynamic and measurement models were assumed to be linear and Gaussian. As a result, the posterior of the state $p(x_k | y_k)$ was derived as a linear transformation of the mean and covariance. For the bootstrap particle filter, the posterior of the state is formulated as a Monte Carlo approximation, with $x_k^{(i)}$ denoting a randomly sampled state for N total particles

$$p(x_k | y_k) \approx \frac{\sum_{i=1}^N p(y_k | x_k^{(i)} = x_k) p(x_k^{(i)} = x_k)}{\sum_{j=1}^N p(y_k | x_k^{(j)})}. \tag{2.11}$$

The steps to implement to formulate the bootstrap particle filter iteratively can be summarized in the following algorithm [22] [23] [24]

1. Sample N particles from the dynamic function

$$x_k^{(i)} \sim p(x_k | x_{k-1}^{(i)}, u_{k-1}) \quad i = 1, \dots, N. \tag{2.12}$$

2. Update weights from the measurement

$$\begin{aligned} w_k^{*(i)} &\propto p(y_k | x_k^{(i)}) \quad i = 1, \dots, N \\ w_k^{(i)} &= \frac{w_k^{*(i)}}{\sum_{j=1}^N w_k^{*(j)}} \quad i = 1, \dots, N. \end{aligned} \quad (2.13)$$

3. Systematically resample particles from a distribution proportional to the weights

$$\begin{aligned} u_1 &\sim \mathbb{U} \left[0, \frac{1}{N} \right], \\ u_r &= u_1 + \frac{r-1}{N} \quad r = 2, \dots, N, \\ x_k^{(i)} &= \begin{cases} x_k^{(r)} & \text{if } \sum_{j=1}^{i-1} w_k^{(j)} < u_r \leq \sum_{j=1}^i w_k^{(j)} \end{cases} \quad i = 1, \dots, N, \\ w_k^{(i)} &= \frac{1}{N} \quad i = 1, \dots, N. \end{aligned} \quad (2.14)$$

2.3.3 Observability in Nonlinear Discrete-Time State-Space Models

Analyzing observability in linear and nonlinear systems assesses how well the states of the model can be inferred from the observations. The observability gramian W_O has traditionally been used in model reduction of linear dynamic systems, but empirical methods have been adapted to accommodate nonlinear systems [25]. The observability gramian provides insight into the states with the most influence on the output by analyzing the rank condition of the observability gramian matrix, which in return allows for analytically evaluating observability.

To empirically calculate the observability gramian, 2^n unique combinations of the initial state $x_0 \in \mathbb{R}^n$ are generated, with n denoting the numbers of states [20] [25] [26]. The initial condition x_0 will be assigned as perturbations $\delta_i \in \mathbb{R}^n$ around a nominal value $x_{nom} \in \mathbb{R}^n$

$$\begin{aligned} x_{0,i} &= x_{nom} + \delta_i, \\ \delta_i &= c \cdot S \cdot t_i, \quad \text{with } S = \text{diag}(x_{\max}). \end{aligned} \quad (2.15)$$

The scalar constant c corresponds to the magnitude of deviation, $S \in \mathbb{R}^{n,n}$ is the diagonal matrix of the maximum of each state and parameter. The row

vector $t_i \in R^n$ is the specified perturbation i from the orthonormal matrix

$$\begin{aligned}
 T &= [t_1 \ t_2 \ \dots \ t_{2^n}] \in R^{n,2^n} \\
 t_1^T &= \frac{1}{\sqrt{2^n}} [-1 \ -1 \ \dots \ -1] \\
 t_2^T &= \frac{1}{\sqrt{2^n}} [-1 \ -1 \ \dots \ 1] \\
 &\vdots \\
 t_{2^n}^T &= \frac{1}{\sqrt{2^n}} [1 \ 1 \ \dots \ 1].
 \end{aligned} \tag{2.16}$$

After calculating the 2^n perturbations of the initial state, the output $y_{1:K,i}$ can be derived sequentially with Equations (2.3) from time steps $k = 1, \dots, K$. Then, the empirical observability with discrete time steps $t = k \, dt$ can be expressed as such

$$W_O = (cS)^{-1} \sum_{k=1}^K T \Psi[k] T^T dt (cS)^{-1}, \tag{2.17}$$

with the positive definite matrix $\Psi[k]$ being assigned as the following

$$\Psi_{i,j}[k] = |y_i[k] - y_0[k]| |y_j[k] - y_0[k]|. \tag{2.18}$$

If the states have varying magnitudes of scale, it is recommended [20] to rerun the observability gramian algorithm again with the following transformations

$$x_s = \sqrt{\text{diag}(W_O)}, \quad T_s = \text{diag}^{-1}(x_s). \tag{2.19}$$

The new state perturbations are assigned as the transformation $\bar{x} = T_s x$. To make inferences about of the observability gramian, one can check the rank condition of the matrix by analyzing the nonzero singular values of W_0 . If there exists singular values near zero (e.g. < 0.1), a subset of the dynamics of the system can not be observed from the measurements.

2.3.4 Related Work

Nonlinear state estimation of the **ICING** model has been investigated in a previous study by Aguirre-Zapata et al. [19]. The study incorporated an additional measurement and dynamic due to their clinical facility having a

continuous glucose machine that can measure the patient's interstitial glucose, but the formulation of patient insulin-glucose dynamics otherwise remained the same. In the methodology of the paper, all tests regarding estimation performance are tested on deviations from one synthetic patient, with each deviation being assigned the same exogenous insulin input per deviation, and food intake is generally not considered. Overall, their synthetic patients do not constitute a reasonable patient model in comparison to real patients.

The study includes three state estimators: the extended Kalman filter, the unscented Kalman filter, and the particle filter. The majority of the study uses discrete time steps of 5 minutes. While the dynamics of patients develop relatively slowly, such a large time step can introduce approximation errors in the discretization of the ICING model. Another notable design choice is their particle filter containing 5000 total particles. Further analysis of various sized particle filters would motivate the overall trade-off estimation performance compared to the time complexity. State estimation remained in the scope of dynamics represented in the ICING model and an additional interstitial glucose dynamic. As a result, there exists further development of state estimators to predict unrepresented ICING model parameters as well.

2.4 Parameter Estimation

As previously mentioned, there exists unrepresented ICING model parameters. Additionally, these parameters contain time-varying characteristics. The whole-body insulin sensitivity S_I or the endogenous insulin u_{en} dynamically fluctuate over time depending on the patient's metabolic condition. As a result, naively assigning either as a general population constant can hinder the performance of the estimators. The unknown time-varying parameters of interest will be denoted as θ in preparation for the next section.

2.4.1 State Augmentation

A practical method to handle unknown parameters is to augment them to the existing state variables $\tilde{x}_k = [x_k, \theta_k]^\top$, such that Equation (2.3) can be redefined as

$$\begin{aligned}\tilde{x}_k &\sim p(\tilde{x}_k \mid \tilde{x}_{k-1}, u_{k-1}), \\ y_k &\sim p(y_k \mid \tilde{x}_k).\end{aligned}\tag{2.20}$$

An important distinction in Equation (2.20) is that the unknown parameters are treated as stochastic variables with singularity in the process noise. This

can be problematic as all models often have approximation errors, and this can drive the extended Kalman filter to ultimately diverge in the Gaussian approximation, as well as introduce sample impoverishment in particle filters [22].

One method to compensate for singularity is to also add artificial process noise ϵ_k to the unknown parameters to allow for random walking in the state-space, thus, accounting for errors in the dynamic function

$$\theta_k = \theta_{k-1} + \epsilon_{k-1}, \quad \text{with } \epsilon_{k-1} \sim \mathcal{N}(0, W_{k-1}). \quad (2.21)$$

With this method, another unknown parameter W_k is inherently added, but such a parameter can be tuned with prior knowledge of the parameters θ_k .

It should be noted that for state augmentation, the dimension of the state-space increases by the number of unknown parameters. In the case of the particle filter, the number of particles will need to be adjusted to maintain accuracy. A simple heuristic, instead of analytically deriving the exact dependency, could be to have the base number of particles P increase by a common factor for each dimension d , such that the total number of particles needed N is

$$N = P \times 2^d. \quad (2.22)$$

2.4.2 Related Work

The Integral-based Parameter Identification (**IPI**) method is a nonlinear and nonconvex parameter identification method that estimates patient-specific parameters as a least squares solution [18]. By linearly interpolating between the blood glucose measurements in times $[t_0, t_1]$,

$$\tilde{G}(t) = \frac{G(t_1) - G(t_0)}{t_1 - t_0}(t - t_0) + G(t_0), \quad (2.23)$$

the Equation (2.1a) can be rearranged as such for a piecewise constant S_I

$$\begin{aligned} \int_{t_0}^{t_1} \dot{G} \, dt &\approx -p_G \int_{t_0}^{t_1} \tilde{G}(t) \, dt - S_I \int_{t_0}^{t_1} \tilde{G}(t) \frac{Q(t)}{1 + \alpha_G Q(t)} \, dt \\ &\quad + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G \, dt \\ G(t_1) - G(t_0) &\approx -p_G \int_{t_0}^{t_1} \tilde{G}(t) \, dt - S_I \int_{t_0}^{t_1} \tilde{G}(t) \frac{Q(t)}{1 + \alpha_G Q(t)} \, dt \end{aligned} \quad (2.24)$$

$$+ \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G dt, \quad (2.25)$$

and S_I can be expressed as a linear equation

$$\underbrace{\int_{t_0}^{t_1} \tilde{G}(t) \frac{Q(t)}{1 + \alpha_G Q(t)} dt}_{A(t_0, t_1)} S_I \approx \underbrace{\left(G(t_0) - G(t_1) - p_G \int_{t_0}^{t_1} \tilde{G}(t) dt + \frac{1}{V_G} \int_{t_0}^{t_1} (P(t) + E_G - C_G) dt \right)}_{b(t_0, t_1)}. \quad (2.26)$$

It is often the case with longer measurement periods ($t_1 - t_0$), that identifying multiple S_I values within the time period may be necessary. For instance, if every 120 minutes glucose measurements are taken from the patient, but every 60 minutes a patient's insulin sensitivity should be identified $S_I \in \mathbb{R}^2$. Both insulin sensitivity parameters at 60 minutes S_I^{60} and 120 minutes S_I^{120} can be derived as the least squares solution from the system of linear equations

$$\begin{bmatrix} A(t_0, t_0 + 30) & 0 \\ A(t_0 + 30, t_0 + 60) & 0 \\ 0 & A(t_0 + 60, t_0 + 90) \\ 0 & A(t_0 + 90, t_0 + 120) \end{bmatrix} \begin{bmatrix} S_I^{60} \\ S_I^{120} \end{bmatrix} = \begin{bmatrix} b(t_0, t_0 + 30) \\ b(t_0 + 30, t_0 + 60) \\ b(t_0 + 60, t_0 + 90) \\ b(t_0 + 90, t_0 + 120) \end{bmatrix}. \quad (2.27)$$

The fundamental flaw of the **IPI** method is the assumption of linear blood glucose dynamics between measurements of times $[t_0, t_1]$. If we closely analyze the Figure 2.1, linearly interpolating blood glucose introduces residuals, due to fitting errors. Consequently, the residuals will propagate into the integrals in the Equation (2.26), resulting in errors in the parameter calculation. Other methods not relying on linearization of the measurements must be explored to allow for realistic estimation of parameters, thus giving better insight of the patient's health status to clinicians.

2.5 Summary

The **ICING** model provides a foundation to model glucose removal, insulin pharmacokinetics, and gastric absorption of glucose for tight glucose control in critically ill patients. Utilizing nonlinear state estimators such as the extended Kalman filter and the bootstrap particle filter allows for

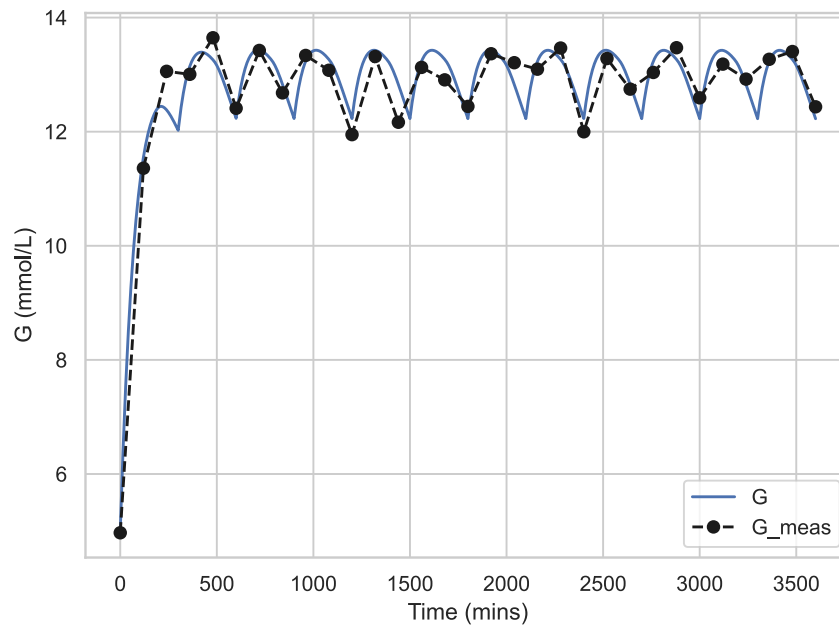


Figure 2.1: Blood glucose and its corresponding measurements from an example patient trajectory from the **ICING** model.

optimal estimation of the states over time with consideration of model approximation errors and measurement noise. Consideration of realistic synthetic patients and systematically motivated parameterization of filters will improve generalizability of the estimators. The current method of **IPI** is formulated with strict assumptions of blood glucose measurements, which limits the ability for clinicians to infer the health of the patient from the approximated patient-specific parameters. More in-depth parameter estimation methods can improve model identifiability for glucoregulation in critically ill patients.

Chapter 3

Methodology

The necessary methodology to achieve the goals introduced in Section 1.4 will be presented in this chapter. It begins with the design of patient trajectories to simulate critically ill patients in this paper. The observability of the ICING model is analyzed to learn if the states can be inferred from the measurements. Based on the observed performance, nonlinear state estimators are utilized to assess the model's ability to estimate unmeasured dynamics. Finally, the parameter estimation methods used to estimate unknown constant and time-varying insulin sensitivity are introduced.

3.1 Synthetic Patient Trajectories

3.1.1 Initial States, Inputs, and Insulin Sensitivity

Synthetic patient trajectories will be simulated to model clinical insulin-glucose data observed in patients in the intensive care unit. A patient trajectory consists of simulated insulin-glucose dynamics (G , Q , I , P_1 , P_2 , u_{en} , P) represented in Equations (2.1a) to (2.1g) corrupted with additive white gaussian process noise q_x , known insulin sensitivity S_I , exogenous insulin u_{ex} , enteral feed D , and parenteral feed P_N . The population constants listed in Table 2.1 are assigned to the constant parameters for all trajectories. To introduce variance between patient trajectories, each trajectory has unique randomly sampled initial inputs observed from doses given to patients in the Karolinska University Hospital and a randomly assigned insulin sensitivity.

The GLUC-ICU dataset that will be made use of in this thesis consists of 27113 patients in the intensive care unit at Karolinska University hospital. 60.5% of the total number of patients are not given insulin, due to patients

being in a stable enough condition or otherwise. From the 10702 patients that intake insulin at some period during their stay in critical care, there exists 100 million minutes of feeding data and 80 million minutes of insulin data to sample from. Patients in the dataset were given discrete inputs of enteral, parenteral, and exogenous insulin. A joint histogram for enteral and parenteral feeding, as well as a histogram for exogenous insulin were constructed. Subsequently, all inputs can be sampled from their respective histogram.

The insulin sensitivity has been shown to follow a log-normal distribution [27]. Sampling the insulin sensitivity from a log-normal is important to adhere to the physiological constraints typically observed in patients, as the insulin sensitivity is small in magnitude, and sampling from a normal distribution could shift the insulin sensitivity to negative or too large in magnitude, which can possibly result in poor estimation. The values chosen within this study will be drawn from the distribution^{*}

$$S_{I,0} = \exp(X), \quad \text{with } X \sim \mathcal{N}(-8.05, 0.70^2). \quad (3.1)$$

Each synthetic patient needs to be assigned a unique initial state[†] for the filter

$$x_0 = [G_0 \quad I_0 \quad Q_0 \quad P_{1,0} \quad P_{2,0} \quad u_{en,0}]. \quad (3.2)$$

If it is assumed that the patient begins at a steady state, it is possible to derive the initial states by solving for the roots of the differential equations (2.1a) to (2.1f). This can be computed with the python solver function `scipy.optimize.fsolve`[‡] for each patient in the cohort with the sampled initial inputs and insulin sensitivity

$$x_0 \in \{x_0^* : f(x_0^*, S_{I,0}, D_0, P_{N,0}, u_{ex,0}) = 0\}. \quad (3.3)$$

^{*}These parameters are approximated according to previous work from Davidson et al. [27]

[†]The glucose appearance in the blood P is not included as an initial state in the state-space model. This is a result of patients not reaching the maximum glucose out flux from the gut $P_{\max}$ at the beginning of the simulation. Thus, the dynamic simplifies to a deterministic equation.

[‡]<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.fsolve.html>

3.1.2 Generating Synthetic Patient Trajectories

To ensure generating clinically relevant synthetic patient data, the patient inputs should be chosen to fluctuate over time. The parenteral glucose and insulin inputs were designed such that they are periodic, exponentially decreasing ramps to values half of the initial $P_{N,0}$ and $u_{ex,0}$, over a period of 4 hours, as well as a constant enteral glucose rate

$$\begin{aligned} D_k &= D_0, \\ P_{N,k} &= P_{N,0} \exp\left(-\frac{k \bmod 240}{\ln(2) \times 240}\right), \\ u_{ex,k} &= u_{ex,0} \exp\left(-\frac{k \bmod 240}{\ln(2) \times 240}\right). \end{aligned} \quad (3.4)$$

Two different insulin sensitivity models are considered in this thesis. A constant insulin sensitivity is assigned as a function of the initial sampled S_I

$$S_{I,k} = S_{I,0}, \quad (3.5)$$

and a piecewise insulin sensitivity that consists of linearly changing S_I with intervals of 20 hours (1200 discrete time steps)

$$\begin{aligned} S_{I,1200} &= z_0 S_{I,0} & z_0 &\sim \mathbb{U}[0.5, 2], \\ S_{I,2400} &= z_1 S_{I,1200} & z_1 &\sim \mathbb{U}[0.5, 2], \\ S_{I,3600} &= z_2 S_{I,2400} & z_2 &\sim \mathbb{U}[0.5, 2], \\ S_{I,k} &= \begin{cases} \frac{S_{I,1200} - S_{I,0}}{1200}k + S_{I,0}, & k < 1200, \\ \frac{S_{I,2400} - S_{I,1200}}{1200}(k - 1200) + S_{I,1200}, & 1200 \leq k < 2400, \\ \frac{S_{I,3600} - S_{I,2400}}{1200}(k - 2400) + S_{I,2400}, & 2400 \leq k \leq 3600 \end{cases} \end{aligned} \quad (3.6)$$

To ensure the insulin sensitivity is constrained to physiologically observed values, $S_{I,1200}$, $S_{I,2400}$, $S_{I,3600}$ will be clipped to the minimum and maximum insulin sensitivity observed in the initial insulin sensitivity histogram.

The process noise for the **ICING** model will be assigned with the measurement sampling frequency in mind. Every 2 hours, a blood glucose sample of the patient is measured, resulting in 120 discrete time steps of the ICING model for a step size of $\Delta t = 1$ min. The process noise will be chosen as 10% of the variance of the derived initial states, to signify variation of the true state compared to the model, as well as be scaled by the sampling

frequency

$$q_k \sim \mathcal{N}(0, \text{diag}(\sigma^2)),$$

$$\sigma^2 = \frac{1}{120} \begin{bmatrix} \frac{\sigma_G^2}{10} & \frac{\sigma_Q^2}{10} & \dots & \frac{\sigma_{P_2}^2}{10} & 1.0 \times 10^{-8} \end{bmatrix}^\top. \quad (3.7)$$

Thus, given all the initial states, known insulin sensitivity, and the patient inputs, the patient's insulin-glucose states can be derived dynamically as shown in Figure 3.1, according to the discretized **ICING** model with dynamics of the form as shown in Equation (2.2).

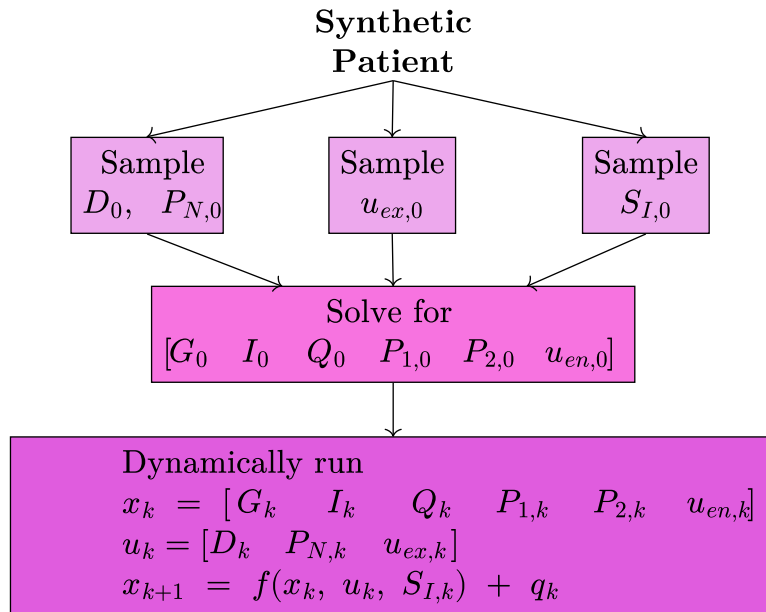


Figure 3.1: Process of generating synthetic patients.

[‡]The process noise for endogenous insulin u_{en} was chosen as a relatively small value due to plasma insulin already being modeled with uncertainty.

3.2 Observability of the Intensive Control Insulin-Nutrition-Glucose Model

In Section 3.1, synthetic patient trajectories were created by first randomly sampling initial states, and then generating the 60 hours of patient's insulin-glucose states for a constant insulin sensitivity according to the ICING model. For the ICING model to be classified as observable, it must be possible to infer the insulin-glucose states, given a sufficient number of observations. To establish if the formulation of the ICING model is observable, a mathematical analysis can determine the relationship of the insulin-glucose states and the observations as previously discussed in Section 2.3.3.

250 patients will be randomly sampled to investigate the observability of the ICING model across various permutations of each random nominal state vector. The maximum value of each state x_{max} will be chosen from the patient trajectories found in Section 3.1

$$S = \text{diag}(x_{max}), \quad (3.8)$$

as well with a maximum perturbation constant of $c = 0.1$. With the following defined inputs, the observability gramian W_O can be derived to analyze the singular values of all the calculated matrices using single value decomposition. To also analyze which states are the most observable respectively to each other, the diagonals of the observability gramian can be used to propose that diagonal entries greater in magnitude provide more information in relation to the observations. Formulating an approximation of the observability will provide insight into the performance of the estimators in the following section.

3.3 Nonlinear State Estimation

The extended Kalman filter and the particle filter are two different nonlinear state estimation methods presented in Section 2.3. To compare each estimator with the ICING model, the total time to run each filter and the mean squared error (MSE) of the blood glucose concentration estimates will be recorded for 250 synthetic patient trajectories. The simulations will span 12 hours to allow each estimator to converge, with glucose measurements corrupted with additive Gaussian noise ($\mathcal{N}(0, 0.25^2)$).

The extended Kalman filter parameters will be defined as

1. Initial covariance matrix of

$$P_0 = \text{diag} \left(\left[\sigma_G^2 \quad \sigma_Q^2 \quad \cdots \quad \sigma_{u_{en}}^2 \right]^T \right). \quad (3.9)$$

2. Process noise covariance matrix of

$$Q = \text{diag}(\sigma^2). \quad (3.10)$$

3. Measurement noise covariance matrix of

$$R = [0.25^2]. \quad (3.11)$$

Choosing the initial covariance matrix larger than the process noise covariance matrix is chosen to allow the filter to have large uncertainty at the beginning of the simulation until the first measurement.

Particle filters containing (100, 250, 500, 1000, 1500, 3000, 5000) particles will be used to compare the relationship of the estimation performance over an increasing number of particles. The notation of the extended Kalman filter will be abbreviated as EKF and for a particle filter with N particles, it will be represented as PF N. The initial particles for each particle filter will be randomly sampled from the derived initial state histograms in Section 3.1, compared to the extended Kalman filter that assigns the initial states from the mean of the fitted Gaussian distributions.

The error of the estimators for a patient i will be represented as the **MSE** of the true blood glucose concentration sequence G and an estimated blood glucose concentration sequence $\hat{G}$ of length L

$$\text{MSE}_i = \frac{1}{L} \sum_{j=1}^L \left(G_i[j] - \hat{G}_i[j] \right)^2. \quad (3.12)$$

A Pareto curve can be utilized to visualize the "optimal" number of particles in terms of the trade-off between time complexity and error. The resulting number of particles needed per dimension will be calculated to gauge the rough number of particles needed to include additional parameters in the state-space in the following section.

3.4 Parameter Estimation

The **IPI** method introduced in Section 2.4.2 will be the baseline implementation of estimating insulin sensitivity in patients and will be compared against the extended Kalman filter and the particle filter with state augmented filters. Each method will estimate a constant insulin sensitivity and a piecewise insulin sensitivity to draw conclusions on the limitations of each estimator. The simulation will last for 60 hours, as changes in insulin sensitivity are relatively slow in time. There will be a total of 250 synthetic patient trajectories.

Due to the insulin sensitivity being log-normal as previously explained in Section 3.1.1, the artificial process noise $q_{S_I,k}$ needed to model insulin sensitivity as a random walk in the dynamic model is multiplicative, not additive. The process noise distribution is chosen as

$$q_{S_I,k} \sim \mathcal{N}\left(1, \frac{\sigma_{S_I}^2}{1200}\right), \quad \text{with } \sigma_{S_I} = \frac{1}{3} \quad (3.13)$$

to model the random walk of insulin sensitivity with a coefficient of variation of $\frac{1}{3}$ over a period over 20 hours.

The extended Kalman filter configuration for this section is the following

1. Initial covariance matrix of

$$\hat{P}_0 = \text{diag}\left(\left[\left(\frac{\text{med}(G)}{10}\right)^2 \quad \dots \quad \left(\frac{\text{med}(u_{en})}{10}\right)^2 \quad 1 \times 10^{-3}\right]^T\right). \quad (3.14)$$

2. Process noise matrix of

$$\hat{Q} = \text{diag}\left(\left[\begin{array}{c} \sigma^2 \\ \frac{\sigma_{S_I}^2}{1200} \end{array}\right]\right). \quad (3.15)$$

3. Measurement noise of R

The initial variance of the insulin sensitivity is chosen relatively high for the extended Kalman filter to account for the error in the Gaussian transformation of the log process noise. Like the previous section, the particle filter will sample initial particles directly from the histograms in Section 3.1, and the extended Kalman filter assigns the initial states from the mean of all states and insulin sensitivity.

Assessing the estimation of the insulin sensitivity will be evaluated using the **MSE** of the true insulin sensitivity S_I and the estimated insulin sensitivity

$\hat{S}_I$ for a given patient i , with a total length of L

$$\text{MSE}_{S_I,i} = \frac{1}{L} \sum_{j=1}^L \left(S_{Ii}[j] - \hat{S}_{Ii}[j] \right)^2. \quad (3.16)$$

In the next chapter, the results of the synthetic patient cohort generation, observability, state estimation, and parameter estimation will be presented and analyzed.

Chapter 4

Results

4.1 Synthetic Patient Trajectories

Approximating probability mass functions was used to analyze the characteristics of the target population of critically ill patients, with inputs sampled directly from clinical data from the Karolinska University Hospital. Figure 4.1 is divided into two subfigures, with Figure 4.1a consisting of the histograms of the inputs u_{ex} , D , and P_N , as well as the insulin sensitivity parameter S_I , and Figure 4.1b is visualizing the histograms of the derived initial states G , I , Q , P_1 , P_2 , P , u_{en} , with each state having the following means and standard deviations in Table 4.1.

Table 4.1: Mean and standard deviation of the initial states.

	G	Q	I	P_1	P_2	u_{en}
	mmol/L	mU/min	mU/min	mmol	mmol	mU/min
μ	9.67	41.8	61.4	9.13	45.6	41.9
σ	4.47	69.9	59.0	10.2	51.3	3.76

Sampling numerous synthetic patients presented a diverse distribution of initial states from the randomly assigned inputs and insulin sensitivity. This allowed for sampling of many unique variations of the patient's insulin-glucose states over time without relying on manual interventions. Previous work in synthetic patient cohorts in glycaemic control was often limited to perturbations in previously recorded clinical data [16] or from integral-fitting [17] [18], but in this approach, a clear computational method towards reproducible generation of patient data was made feasible to analyze and

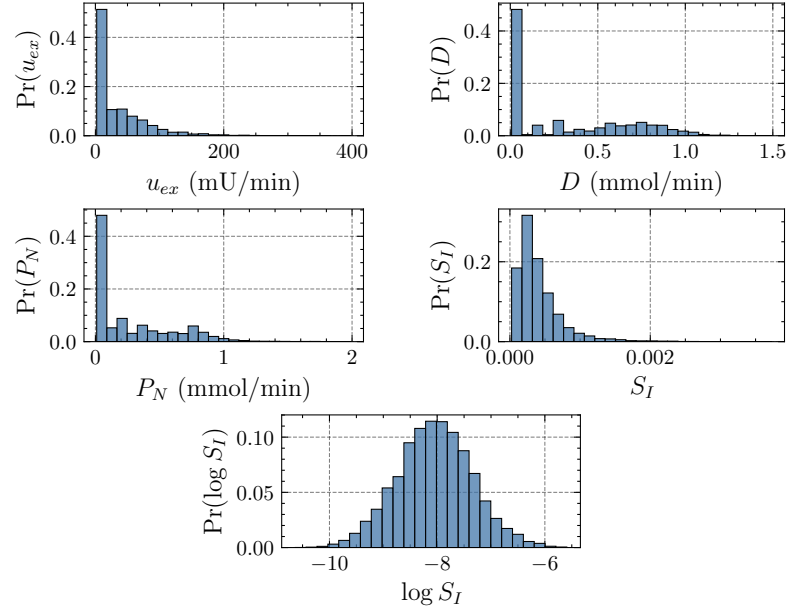
compare the methods in the following sections.

4.2 Observability of the Intensive Glucose Insulin-Nutrition-Glucose Model

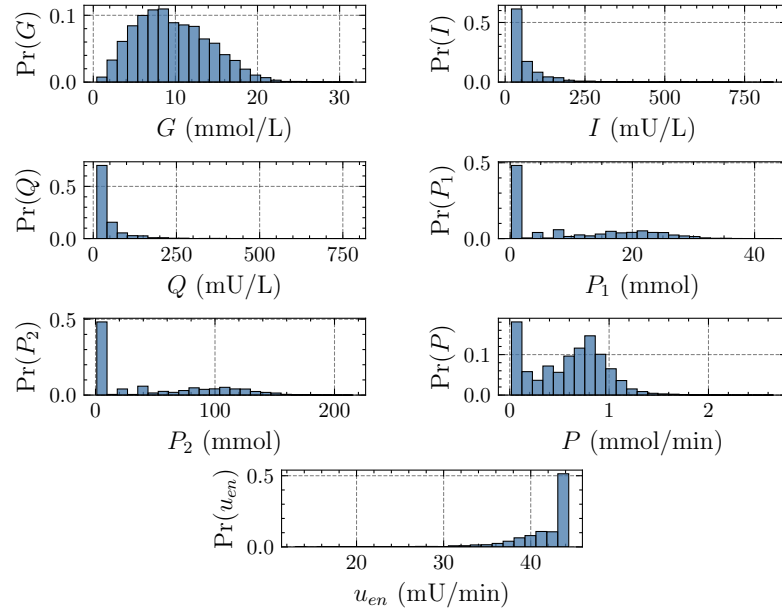
To investigate the observability of the **ICING** model, the eigenvalues and diagonals from the observability gramians were computed. In Figure 4.2a, a histogram of eigenvalues from the 250 calculated empirical observability gramians were plotted, and were transformed to a log-scale to easily visualize the spread of values. The distribution of the eigenvalues spans across 7 orders of magnitudes, with three distinct peaks in the histogram. In the other Figure 4.2b, a box plot for each dynamic in the empirical observability gramian is plotted to show the respective locality of the diagonals $\text{diag}(W_O)$ over all patients. As a result of scaling all dynamics in the empirical observability gramian, the diagonal values are comparable amongst the dynamics.

The blood glucose concentration G is the most observable among all states. This is to be expected as the observations in our state-space model are noisy blood glucose measurements. The blood glucose appearance in the stomach and the endogenous insulin are the next observable states, with a median roughly of 105 and 96.1, as well as an approximate interquartile range of 160 and 147 respectively. In Equation (2.1a), the change in G is dependent on a constant consisting of a scaled multiple of the total glucose appearance P , which is a function of both the glucose appearance in the stomach and gut P_1, P_2 . The diagonals suggest that perturbations in the amount of glucose in the stomach have more influence on the blood glucose output compared to the glucose in the gut. Blood glucose measurements on their own provide minimal information about interstitial and plasma insulin production, as seen in the lower gramian values.

The observability analysis suggests that given all blood glucose measurements, one could not infer all insulin-glucose states with the **ICING** model. One would have issues with decoupling the insulin dynamics, as well with the glucose absorption states. While this thesis focuses on model performance, specifically for estimating blood glucose, the observability analysis highlights that having no plasma insulin measurements is a bottleneck, and the overall complexity of attempting to model a patient's insulin-glucose system.



(a) Initial inputs and parameters



(b) Initial States

Figure 4.1: Histograms of the characteristics of the synthetic patients.

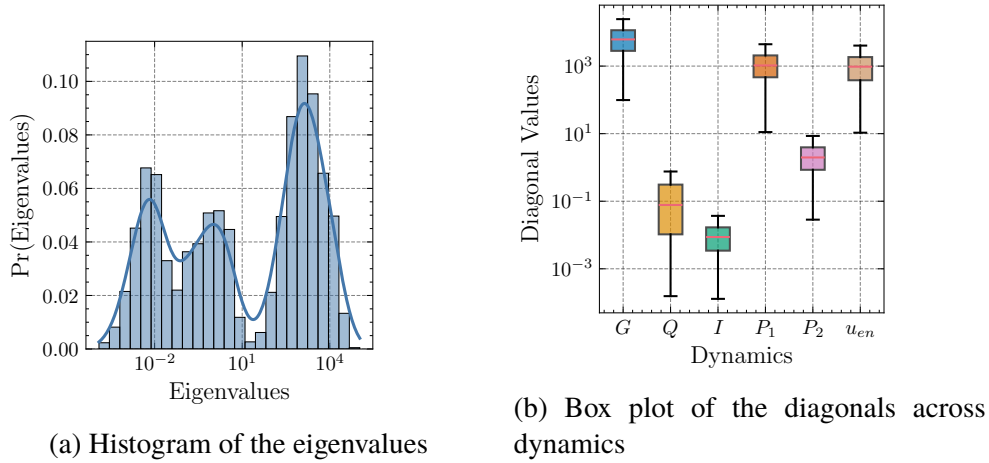


Figure 4.2: Analysis across all calculated empirical observability gramians.

4.3 Nonlinear State Estimation

For each filter, the **MSE** is averaged among all patients in Table 4.2, as well as the standard deviation SD of the error. The extended Kalman filter overall performs worse compared to the particle filters. Improvements in performance compared to the extended Kalman filter can be regarded in a particle filter with just 100 particles, highlighting the particle filter's ability to adapt quickly to errors in the state estimates when resampling after the initial measurement. With each increment of particles, each filter decreases the error of the estimate, as well as the spread of errors. It is evident that the greater number of particles a particle filter has, the better the filter can approximate the posterior distribution of the states given the measurements, resulting in a more accurate estimate of blood glucose.

To compare the overall performance of the filters, it is helpful to analyze Figure 4.3. The x -axis is the average total time to run each estimator. The y -axis is the average **MSE** for all deviations of the initial states, plotted in the log-domain. In terms of error, there is a plateau in improvements in performance after 1000 particles. The "optimum" number of particles was chosen as 1000 particles, as there is a balance of adequate performance and lower runtimes. For the next section, the insulin sensitivity will be included in the state-space. Consequently, the state-space increases to 7. To have a rough range of the number of particles needed to accommodate a larger dimension in the state-space, the approximate number of particles is chosen as 2000, according to Section 2.4.1.

Table 4.2: Means and standard deviations of MSE for each estimators.

Estimator	Mean MSE	SD MSE
EKF	3.46	9.64
PF 100	1.58	5.28
PF 250	1.26	4.26
PF 500	1.04	3.25
PF 1000	0.926	2.55
PF 1500	0.907	2.48
PF 2500	0.865	2.33
PF 5000	0.850	2.26

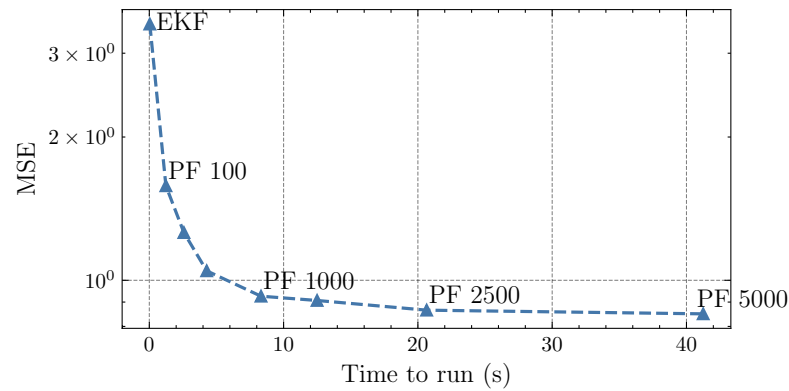


Figure 4.3: Comparison of the total average run time and error across all estimators.

The chosen "optimal" number of particles is much less computationally expensive compared to previous work [19], which used 5000 particles for a state-space with the same number of dimensions. Although the Aguirre et al. study did contain nongaussian measurement noise, the number of particles to estimate the posterior distribution with a known measurement model prior likely does not require that many particles, even if an exponential heuristic was chosen for the particle calculation. If we decided to choose 5000 particles, there would likely be little marginal gain in the performance and would result in 5 times the computational complexity. For a system that may require an estimator to run in approximately real-time, an estimator with 5000 particles would require more computational resources. Hence, our approach is much more practical in clinical settings. If very limited computational resources are allotted, smaller sized particle filters can be utilized instead.

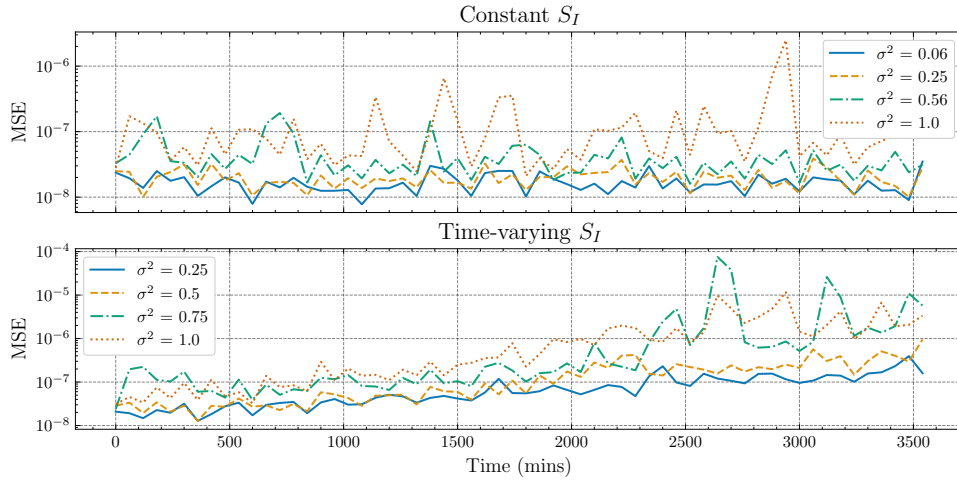


Figure 4.4: Integral-based method insulin sensitivity estimation performance across varying measurement noise variances.

4.4 Parameter Estimation

State augmentation of insulin sensitivity in the extended Kalman filter and the particle filter was implemented to compare the estimation performance of the existing state estimators against the **IPI** method. Both constant and time-varying insulin-sensitivity were used to investigate how characteristics of the parameter affect estimation. Figures 4.4 and 4.5 display the performance of each method according to the time dependency of the parameter. MSE in these figures depicts the mean error for each time step, not the length of the sequence.

The **IPI** method in Figure 4.4 illustrates how the estimation accuracy degrades drastically if any measurement noise is present in the calculation. Therefore, the performance is heavily dependent on the quality of the measurements. Any estimation of the insulin sensitivity from the **IPI** method does not provide any information on the insulin sensitivity, as the output never converges towards the true value(s) in a practical setting with noisy measurements. For this reason, the **IPI** method is removed from the comparison of the extended Kalman filter and particle filter in the following section.

In Figure 4.5, both the extended Kalman filter and the particle filter converge towards the true constant insulin sensitivity. The extended Kalman filter on average plateaus in the estimation of the insulin sensitivity around an MSE of 5^{-7} , while the particle filter with 2000 particles on average roughly

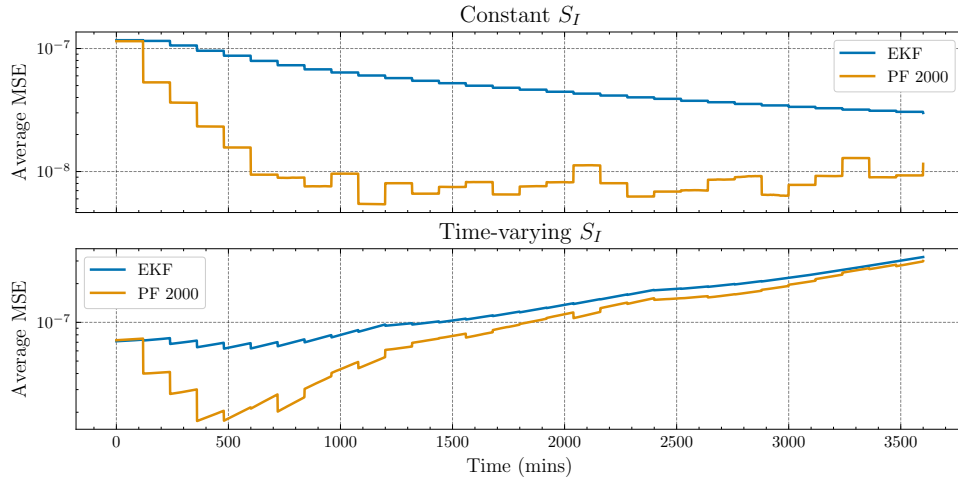


Figure 4.5: Mean MSE over time in parameter estimation methods estimating insulin sensitivity.

converges closer to the true value with a MSE close to 10^{-8} . The extended Kalman filter adapts much more slowly compared to the particle filter as a result of the particles' ability to draw new particles at each measurement, where many Monte Carlo samples can better estimate the insulin-glucose dynamics.

In the time-varying parameter estimation case, it can be seen in Figure 4.5 that both the extended Kalman filter and particle filter struggle to converge. There is a clear lack of ability for the dynamic model to adapt to insulin sensitivity that is changing. Both filters are modeling the piecewise insulin sensitivity with the expected deviation over time $\sigma_{S_I}^2$, but the estimates do not adapt. This can also be observed in the bottom plot in Figure 4.4, as the insulin sensitivity estimates follow a similar trend over time with an increase of error. The observability of the **ICING** model is likely to act as a bottleneck in this instance, as the model would need to decouple the changing insulin sensitivity from the relatively unobservable interstitial and plasma insulin.

Chapter 5

Conclusions

As previously stated, tight glucose control in critically-ill patients has been transitioning from manual ad-hoc sliding protocols to computational methods of control. Evaluation of clinically-validated mechanistic insulin-glucose models with Bayesian filtering has been implemented with unrepresentative synthetic patients and not entirely realistic assumptions of estimating insulin-glucose states in practice. This thesis improved and expanded implementation of tight glucose control by evaluating two different nonlinear state estimators, the Kalman filter and the particle filter, on synthetic patient trajectories generated from nutrition and insulin input distributions from Karolinska University Hospital.

An important contribution this thesis achieved is the framework to sample synthetic patient trajectories. Interventions on unique and diverse synthetic patients are foundational to prepare for real-world application on vulnerable patients. For other researchers with limited access to confidential patient data, a clearly documented and reproducible methodology to generate synthetic patient trajectories is imperative for the continuation of research in tight glucose control. Combining realistic patient data from the intensive care unit along with a clinically-validated patient insulin-glucose model allowed for estimating blood glucose and insulin sensitivity from a variety of synthetic patients.

Estimation of insulin-glucose states and an additional parameter, insulin sensitivity, was implemented on differing Bayesian filters, tying the results of each to the capabilities of the **ICING** model. The experiment replicated real-world circumstances by including uncertainty in the patient model, temporal insulin and nutrition inputs, along with an unknown constant and piecewise insulin sensitivity. This highlighted the need for precise and well-

motivated filter choice, as it was observed that particle filters with 1000 to 5000 particles perform approximately the same. Modeling constant insulin sensitivity was possible by including an additional state in the **ICING** model, and estimates in time steps of 1 minute in the extended Kalman filter and particle filter outperformed the **IPI** method that can only predict every hour in the experiment.

Future work of this thesis can further improve the synthetic patients by including recorded blood glucose or other insulin-glucose states to create a collection of insulin-state distributions not derived analytically. Another consideration for future work of this thesis would be to expand upon the parameter estimation section. This would be to experiment with different dynamic models for insulin sensitivity or look into data-driven methods. Estimation of other parameters like the distribution rate of glucose between the stomach and the gut or the insulin clearance rate in the interstitium could be useful for better patient modeling.

5.1 Ethics and Sustainability

This thesis contributes to the United Nation's *Good Health and Well-being* goal for sustainable development. The outcome of better and efficient control of glucoregulation translates to improved prevention of mortality for children, women, the elderly, and all those who are suffering from traumatic illnesses abroad. Widespread computational control will provide access to potentially lifesaving technology to places in need that have a limited number of clinicians or resources. Contributions in the field of tight glucose control also overlap with diabetes research. As a result, better modeling of the human's insulin-glucose system will improve the research and treatment of type 1 and type 2 diabetes.

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  It has been observed that patients in intensive care are likely to suffer from hyperglycaemia as a result of fluctuating insulin resistance. Controlling blood glucose from insulin and nutrition protocols stemming from hourly glucose measurements has proved to be difficult and not sophisticated enough to consider the dynamic health of the patient. Implementation of tight glucose control has been motivated by the effectiveness of computational estimation of insulin-glucose states in synthetic and real patients by using mechanistic models such as the Intensive Control and Insulin-Nutrition-Glucose (ICING) model. This thesis sought to improve the design of synthetic patients and the application of Bayesian filters for estimation. Better simulation and design of tight glucose controls is crucial to implement improved treatment of patients with critical illnesses. An analytical framework to sample synthetic patient trajectories was proposed that utilized distributions of insulin and enteral/parenteral inputs given to 10702 patients in the Karolinska University Hospital. An estimation of unknown insulin-glucose states from 250 unique patients was compared by analyzing the mean squared error (MSE) and the run time of the estimates in the extended Kalman filter, and a bootstrap particle filter of various Monte Carlo samples for a patient over 12 hours. The extended Kalman filter had an average MSE of 3.46 and a run time of 0.371 seconds, while a particle filter with 1000 Monte Carlo samples achieved an average MSE of 0.926 and a run time of 8.32 seconds. Estimation of an additional parameter of constant insulin sensitivity through state augmentation in another set of 250 synthetic patients resulted in the extended Kalman filter having a MSE of  $5.62 \times 10^{-7}$ , and the particle filter of 1000 particles with a MSE of  $1.49 \times 10^{-7}$ . For a changing insulin sensitivity, both filters had an increase of MSE of one order of magnitude. Given utilizing the ICING model with blood glucose measurements, a particle filter with 1000 particles can estimate blood glucose and constant insulin sensitivity in critically ill patients with a reasonable run time. Further investigation into capturing time-varying insulin sensitivity is needed to better model patient dynamics.
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  "Keywords[eng ]": €€€€
  Bayesian filter, Blood glucose, Critically ill patients, Insulin resistance, Insulin-glucose dynamics, Insulin sensitivity, Model-based control, Tight glycemic control €€€€,
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Det har observerats att patienter inom intensivvården sannolikt drabbas av hyperglykemi till följd av fluktuerande insulinresistens. Att kontrollera blodsockret med insulin- och nutritionsprotokoll som härrör från timvisa glukosmätningar har visat sig vara svårt och inte tillräckligt sofistikerat för att beakta patientens dynamiska hälsotillstånd. Implementeringen av strikt glukoskontroll har motiverats av effektiviteten hos beräkningsbaserad skattnig av insulin-glukostillstånd hos syntetiska och verkliga patienter genom användning av mekanistiska modeller såsom Intensive Control and

Insulin-Nutrition-Glucose (ICING)-modellen. Denna avhandling syftade till att förbättra utformningen av syntetiska patienter och tillämpningen av Bayesianska filter för skattning. Bättre simulering och utformning av strikt glukoskontroll är avgörande för att implementera förbättrad behandling av patienter med kritiska sjukdomstillstånd. Ett analytiskt ramverk för att sampla syntetiska patientförlopp föreslogs, vilket använde fördelningar av insulin- och enterala/parenterala tillförselmängder som gavs till 10702 patienter vid Karolinska University Hospital. En skattning av ökända insulin-glukostillstånd från 250 unika patienter jämfördes genom analys av medelkvadratfelet (MSE) och körtiden för skattningarna i det utvidgade Kalmanfiltret samt ett bootstrap-partikelfilter med olika antal Monte Carlo-samplingar för en patient under 12 timmar. Det utvidgade Kalmanfiltret hade ett genomsnittligt MSE på 3.46 och en körtid på 0.371 sekunder, medan ett partikelfilter med 1000 Monte Carlo-samplingar uppnådde ett genomsnittligt MSE på 0.926 och en körtid på 8.32 sekunder. Skattning av ytterligare en parameter för konstant insulinkänslighet genom tillståndsaugmentering hos ytterligare 250 syntetiska patienter resulterade i att det utvidgade Kalmanfiltret hade ett MSE på 5.62×10^{-7} , och partikelfiltret med 1000 partiklar hade ett MSE på 1.49×10^{-7} . Vid varierande insulinkänslighet hade båda filtren en ökning av MSE med en storleksordning. Givet användningen av ICING-modellen tillsammans med blodsockermätningar kan ett partikelfilter med 1000 partiklar uppskatta blodsocker och konstant insulinkänslighet hos kritiskt sjuka patienter med en rimlig körtid. Ytterligare undersökning av hur tidsvarierande insulinkänslighet kan fångas är nödvändig för att bättre modellera patientdynamiken.

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"Keywords[swe]": €€€€

Bayesiskt filter, Blodglukos, Insulin-glukos dynamiker, Insulinkänslighet, Insulinresisten, Kritiskt sjuka patienter, Modellbaserade kontrollera, Tight glycemic control €€€€,

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